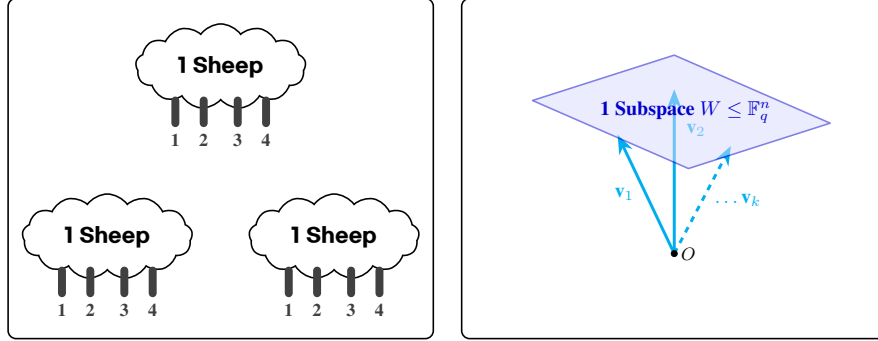


HW#1: Number of Distinct k -dimensional Subspaces in n -dimensional Vector Space

Note. Consider a k -dimensional subspace $W \leq \mathbb{F}_q^n$.



Total legs = (number of sheep) $\times$ (legs per sheep)
Total bases = (number of subspaces) $\times$ (bases per subspace)

We have two steps.

1. Count all ordered k -tuples $(\mathbf{v}_1, \dots, \mathbf{v}_k)$ of linearly independent vectors in V .
2. Divide by the number of ordered bases of one fixed k -dimensional subspace.

Then we have

$$\#\{k\text{-dimensional subspaces of } V\} = \frac{\#\{\text{ordered linearly independent } k\text{-tuples in } V\}}{\#\{\text{ordered bases of one fixed } k\text{-dimensional subspace}\}}.$$

Theorem 1. Let $V = \mathbb{F}_q^n$ be a n -dimensional $\mathbb{F}_2$ -vector space, where q is a prime power. For $0 \leq k \leq n$,

$$\#\{W \leq \mathbb{F}_q^n : \dim W = k\} = \begin{bmatrix} n \\ k \end{bmatrix}_q = \prod_{i=0}^{k-1} \frac{q^n - q^i}{q^k - q^i} = \frac{(q^n - 1)(q^n - q)(q^n - q^2) \cdots (q^n - q^{k-1})}{(q^k - 1)(q^k - q)(q^k - q^2) \cdots (q^k - q^{k-1})}.$$

$$\begin{pmatrix} \boxed{a_{11}} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \times \begin{pmatrix} a_{11} & \boxed{a_{12}} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \times \cdots \times \begin{pmatrix} a_{11} & \cdots & \boxed{a_{1k}} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2k} & \cdots & a_{2n} \\ \vdots & \ddots & a_{3k} & \cdots & \vdots \\ a_{n1} & \cdots & a_{nk} & \cdots & a_{nn} \end{pmatrix} \\ \begin{matrix} \mathbf{v}_1 \neq \mathbf{0} \\ (q^n - 1) \text{ ways} \end{matrix} \times \begin{matrix} \mathbf{v}_2 \notin \text{span}(\mathbf{v}_1) \\ (q^n - q) \text{ ways} \end{matrix} \times \cdots \times \begin{matrix} \mathbf{v}_k \notin \text{span}(\mathbf{v}_i)_{i=1}^{k-1} \\ (q^n - q^{k-1}) \text{ ways} \end{matrix} = \prod_{i=0}^{k-1} (q^n - q^i)$$

$$\begin{pmatrix} \boxed{a_{11}} & a_{12} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix} \times \begin{pmatrix} a_{11} & \boxed{a_{12}} & \cdots & a_{1k} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & a_{kk} \end{pmatrix} \times \cdots \times \begin{pmatrix} a_{11} & a_{12} & \cdots & \boxed{a_{1k}} \\ a_{21} & a_{22} & \cdots & a_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ a_{k1} & a_{k2} & \cdots & \boxed{a_{kk}} \end{pmatrix} \\ \begin{matrix} \mathbf{w}_1 \in W \setminus \{\mathbf{0}\} \\ (q^k - 1) \text{ ways} \end{matrix} \times \begin{matrix} \mathbf{w}_2 \notin \text{span}(\mathbf{v}_1) \\ (q^k - q) \text{ ways} \end{matrix} \times \cdots \times \begin{matrix} \mathbf{w}_k \notin \text{span}(\mathbf{w}_i)_{i=1}^{k-1} \\ (q^k - q^{k-1}) \text{ ways} \end{matrix} = \prod_{i=0}^{k-1} (q^k - q^i)$$